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ORGANIC LIGHT-EMITTING DEVICE

Abstract

An organic light emitting device comprising an anode, a cathode, and a light emitting layer between the anode and the cathode, the light emitting layer including a compound represented by Chemical Formula 1 and a compound represented by Chemical Formula 2, is provided.

##STR00001##

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a National Phase entry pursuant to 35 U.S.C. § 371 of International Application No. PCT/KR2022/006539 filed on May 9, 2022, and claims priority to and the benefit of Korean Patent Application No. 10-2021-0059329 filed on May 7, 2021 and Korean Patent Application No. 10-2022-0056391 filed on May 9, 2022, the disclosures of which are incorporated herein by reference in their entirety.

FIELD OF DISCLOSURE

[0002] The present disclosure relates to an organic light emitting device.

BACKGROUND

[0003] In general, an organic light emitting phenomenon refers to a phenomenon where electric energy is converted into light energy by using an organic material. The organic light emitting device using the organic light emitting phenomenon has characteristics such as a wide viewing angle, an excellent contrast, a fast response time, an excellent luminance, driving voltage and response speed, and thus many studies have proceeded.

[0004] The organic light emitting device generally has a structure which comprises an anode, a cathode, and an organic material layer between the anode and the cathode. The organic material layer frequently has a multilayered structure that comprises different materials in order to enhance efficiency and stability of the organic light emitting device, and for example, the organic material layer may be formed of a hole injection layer, a hole transport layer, a light emitting layer, an electron transport layer, an electron injection layer and the like. In the structure of the organic light emitting device, if a voltage is applied between two electrodes, the holes are injected from an anode into the organic material layer and the electrons are injected from the cathode into the organic material layer, and when the injected holes and electrons meet each other, an exciton is formed, and light is emitted when the exciton falls to a ground state again.

[0005] There is a continuing need for the development of a new material for an organic material used in the organic light emitting device as described above.

RELATED ART

[0006] Korean Unexamined Patent Publication No. 10-2000-0051826

SUMMARY

[0007] It is an object of the present disclosure to provide an organic light emitting device having improved driving voltage, efficiency and lifetime.

[0008] In order to achieve the above object, according to the present disclosure, there is provided an organic light emitting device comprising: [0009] an anode; a cathode; and a light emitting layer between the anode and the cathode, [0010] wherein the light emitting layer includes a compound represented by the following Chemical Formula 1 and a compound represented by the following Chemical Formula 2.

##STR00002## [0011] in Chemical Formula 1, [0012] Y is O or S, [0013] X.sub.1 to X.sub.3 are each independently CH or N, with the proviso that at least one of X.sub.1 to X.sub.3 is N, [0014] Ar.sub.1 and Ar.sub.2 are each independently a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing any one or more selected from the group consisting of N, O and S, [0015] n is an integer of 1 to 6, [0016] R.sub.1 is each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting

of N, O and S, and [0017] Ar.sub.3 is a substituent represented by the following Chemical Formula 1-1 or Chemical Formula 1-2,
##STR00003## [0018] in Chemical Formulas 1-1 and 1-2, [0019] p is an integer of 1 to 8, [0020] q is an integer of 1 to 10, [0021] R.sub.11 and R.sub.12 are each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting of N, O and S, [0022] A is a benzene ring, and [0023] Ar.sub.12 is a substituted or unsubstituted C.sub.6-60 aryl,
##STR00004## [0024] in Chemical Formula 2, [0025] n' and m' are each independently an integer from 1 to 7, [0026] R'.sub.1 and R'.sub.2 are each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting of N, O and S, and [0027] Ar'.sub.1 and Ar'.sub.2 are each independently a substituted or unsubstituted C.sub.6-17 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more heteroatoms selected from the group consisting of N, O and S, [0028] with the proviso that at least one of R'.sub.1 and R'.sub.2 is deuterium; and/or at least one of Ar'.sub.1 and Ar'.sub.2 is substituted with one or more deuteriums. [0029] The above-mentioned organic light emitting device is excellent in driving voltage, efficiency and lifetime.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 shows an example of an organic light emitting device comprising a substrate **1**, an anode **2**, a light emitting layer **3**, and a cathode **4**.

[0031] FIG. 2 shows an example of an organic light emitting device comprising a substrate **1**, an anode **2**, a hole injection layer **5**, a hole transport layer **6**, a light emitting layer **3**, an electron transport layer **7**, an electron injection layer **8** and a cathode **4**.

[0032] FIG. 3 shows an example of an organic light emitting device comprising a substrate **1**, an anode **2**, a hole injection layer **5**, a hole transport layer **6**, an electron blocking layer **9**, a light emitting layer **3**, a hole blocking layer **10**, an electron transport layer **7**, an electron injection layer **8** and a cathode **4**.

DETAILED DESCRIPTION

[0033] Hereinafter, embodiments of the present disclosure will be described in more detail to facilitate understanding of the invention.

[0034] As used herein, the notation

##STR00005##

or  custom-character means a bond linked to another substituent group.

[0035] As used herein, the term “substituted or unsubstituted” means being unsubstituted or substituted with one or more substituents selected from the substituent group consisting of deuterium; a halogen group; a nitrile group; a nitro group; a hydroxy group; a carbonyl group; an ester group; an imide group; an amino group; a phosphine oxide group; an alkoxy group; an aryloxy group; an alkylthioxy group; an arylthioxy group; an alkylsulfoxy group; an arylsulfoxy group; a silyl group; a boron group; an alkyl group; a cycloalkyl group; an alkenyl group; an aryl group; an aralkyl group; an aralkenyl group; an alkylaryl group; an alkylamine group; an aralkylamine group; a heteroarylamine group; an arylamine group; an arylphosphine group; and a heterocyclic group containing one or more of N, O and S atoms, or being unsubstituted or substituted with a substituent from the above substituent group which is further substituted by one or more selected from the above substituent group.

[0036] In the present disclosure, the carbon number of a carbonyl group is not particularly limited, but is preferably 1 to 40. Specifically, the carbonyl group may be a compound having the following

structural formulas, but is not limited thereto.

##STR00006##

[0037] In the present disclosure, an ester group may have a structure in which oxygen of the ester group may be substituted by a straight-chain, branched-chain, or cyclic alkyl group having 1 to 25 carbon atoms, or an aryl group having 6 to 25 carbon atoms. Specifically, the ester group may be a compound having the following structural formulas, but is not limited thereto.

##STR00007##

[0038] In the present disclosure, the carbon number of an imide group is not particularly limited, but is preferably 1 to 25. Specifically, the imide group may be a compound having the following structural formulas, but is not limited thereto.

##STR00008##

[0039] In the present disclosure, a silyl group specifically includes a trimethylsilyl group, a triethylsilyl group, a t-butyldimethylsilyl group, a vinyltrimethylsilyl group, a propyldimethylsilyl group, a triphenylsilyl group, a diphenylsilyl group, a phenylsilyl group and the like, but is not limited thereto.

[0040] In the present disclosure, a boron group specifically includes a trimethylboron group, a triethylboron group, a t-butyldimethylboron group, a triphenylboron group, and a phenylboron group, but is not limited thereto.

[0041] In the present disclosure, examples of a halogen group include fluorine, chlorine, bromine, or iodine.

[0042] In the present disclosure, the alkyl group may be straight-chain or branched-chain, and the carbon number thereof is not particularly limited, but is preferably 1 to 40. According to one embodiment, the carbon number of the alkyl group is 1 to 20. According to another embodiment, the carbon number of the alkyl group is 1 to 10. According to another embodiment, the carbon number of the alkyl group is 1 to 6. Specific examples of the alkyl group include methyl, ethyl, propyl, n-propyl, isopropyl, butyl, n-butyl, isobutyl, tert-butyl, sec-butyl, 1-methyl-butyl, 1-ethyl-butyl, pentyl, n-pentyl, isopentyl, neopentyl, tert-pentyl, hexyl, n-hexyl, 1-methylpentyl, 2-methylpentyl, 4-methyl-2-pentyl, 3,3-dimethylbutyl, 2-ethylbutyl, heptyl, n-heptyl, 1-methylhexyl, cyclopentylmethyl, cyclohexylmethyl, octyl, n-octyl, tert-octyl, 1-methylheptyl, 2-ethylhexyl, 2-propylpentyl, n-nonyl, 2,2-dimethylheptyl, 1-ethyl-propyl, 1,1-dimethyl-propyl, isohexyl, 2-methylpentyl, 4-methylhexyl, 5-methylhexyl, and the like, but are not limited thereto.

[0043] In the present disclosure, the alkenyl group may be straight-chain or branched-chain, and the carbon number thereof is not particularly limited, but is preferably 2 to 40. According to one embodiment, the carbon number of the alkenyl group is 2 to 20. According to another embodiment, the carbon number of the alkenyl group is 2 to 10. According to still another embodiment, the carbon number of the alkenyl group is 2 to 6. Specific examples thereof include vinyl, 1-propenyl, isopropenyl, 1-butenyl, 2-butenyl, 3-butenyl, 1-pentenyl, 2-pentenyl, 3-pentenyl, 3-methyl-1-butenyl, 1,3-butadienyl, allyl, 1-phenylvinyl-1-yl, 2-phenylvinyl-1-yl, 2,2-diphenylvinyl-1-yl, 2-phenyl-2-(naphthyl-1-yl) vinyl-1-yl, 2,2-bis(diphenyl-1-yl) vinyl-1-yl, a stilbenyl group, a styrenyl group, and the like, but are not limited thereto.

[0044] In the present disclosure, a cycloalkyl group is not particularly limited, but the carbon number thereof is preferably 3 to 60. According to one embodiment, the carbon number of the cycloalkyl group is 3 to 30. According to another embodiment, the carbon number of the cycloalkyl group is 3 to 20. According to still another embodiment, the carbon number of the cycloalkyl group is 3 to 6. Specific examples thereof include cyclopropyl, cyclobutyl, cyclopentyl, 3-methylcyclopentyl, 2,3-dimethylcyclopentyl, cyclohexyl, 3-methylcyclohexyl, 4-methylcyclohexyl, 2,3-dimethylcyclohexyl, 3,4,5-trimethylcyclohexyl, 4-tert-butylcyclohexyl, cycloheptyl, cyclooctyl, and the like, but are not limited thereto.

[0045] In the present disclosure, an aryl group is not particularly limited, but the carbon number thereof is preferably 6 to 60, and it may be a monocyclic aryl group or a polycyclic aryl group.

According to one embodiment, the carbon number of the aryl group is 6 to 30. According to one embodiment, the carbon number of the aryl group is 6 to 20. The aryl group may be a phenyl group, a biphenyl group, a terphenyl group or the like as the monocyclic aryl group, but is not limited thereto. The polycyclic aryl group includes a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, a perylenyl group, a chrysenyl group, a fluorenyl group or the like, but is not limited thereto.

[0046] In the present disclosure, the fluorenyl group may be substituted, and two substituents may be linked with each other to form a spiro structure. In the case where the fluorenyl group is substituted.

##STR00009##

and the like can be formed. However, the structure is not limited thereto.

[0047] In the present disclosure, a heterocyclic group is a heterocyclic group containing one or more of O, N, Si and S as a heteroatom, and the carbon number thereof is not particularly limited, but is preferably 2 to 60. Examples of the heterocyclic group include a thiophene group, a furan group, a pyrrole group, an imidazole group, a thiazole group, an oxazol group, an oxadiazol group, a triazol group, a pyridyl group, a bipyridyl group, a pyrimidyl group, a triazine group, an acridyl group, a pyridazine group, a pyrazinyl group, a quinolinyl group, a quinazoline group, a quinoxalinyl group, a phthalazinyl group, a pyridopyrimidinyl group, a pyridopyrazinyl group, a pyrazinopyrazinyl group, an isoquinoline group, an indole group, a carbazole group, a benzoxazole group, a benzoimidazole group, a benzothiazol group, a benzocarbazole group, a benzothiophene group, a dibenzothiophene group, a benzofuranyl group, a phenanthroline group, an isoxazolyl group, a thiadiazolyl group, a phenothiazinyl group, a dibenzofuranyl group, and the like, but are not limited thereto.

[0048] In the present disclosure, the aryl group in the aralkyl group, the aralkenyl group, the alkylaryl group and the arylamine group is the same as the aforementioned examples of the aryl group. In the present disclosure, the alkyl group in the aralkyl group, the alkylaryl group and the alkylamine group is the same as the aforementioned examples of the alkyl group. In the present disclosure, the heteroaryl in the heteroarylamine can be applied to the aforementioned description of the heterocyclic group. In the present disclosure, the alkenyl group in the aralkenyl group is the same as the aforementioned examples of the alkenyl group. In the present disclosure, the aforementioned description of the aryl group may be applied except that the arylene is a divalent group. In the present disclosure, the aforementioned description of the heterocyclic group can be applied except that the heteroarylene is a divalent group. In the present disclosure, the aforementioned description of the aryl group or cycloalkyl group can be applied except that the hydrocarbon ring is not a monovalent group but formed by combining two substituent groups. In the present disclosure, the aforementioned description of the heterocyclic group can be applied, except that the heterocycle is not a monovalent group but formed by combining two substituent groups.

[0049] Hereinafter, the present disclosure will be described in detail for each configuration.

Anode and Cathode

[0050] An anode and a cathode used in the present disclosure mean electrodes used in an organic light emitting device.

[0051] As the anode material, generally, a material having a large work function is preferably used so that holes can be smoothly injected into the organic material layer. Specific examples of the anode material include metals such as vanadium, chrome, copper, zinc, and gold, or an alloy thereof: metal oxides such as zinc oxides, indium oxides, indium tin oxides (ITO), and indium zinc oxides (IZO); a combination of metals and oxides, such as ZnO:Al or SnO.sub.2:Sb; conductive polymers such as poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDOT), polypyrrole, and polyaniline, and the like, but are not limited thereto.

[0052] As the cathode material, generally, a material having a small work function is preferably

used so that electrons can be easily injected into the organic material layer. Specific examples of the cathode material include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin, and lead, or an alloy thereof; a multilayered structure material such as LiF/Al or LiO.sub.2/Al, and the like, but are not limited thereto.

Light Emitting Layer

[0053] The light emitting layer used in the present disclosure is a layer that can emit light in the visible light region by combining holes and electrons transported from the anode and the cathode. Generally, the light emitting layer includes a host material and a dopant material, and in the present disclosure, the compound represented by Chemical Formula 1 and the compound represented by Chemical Formula 2 are included as a host.

[0054] In one embodiment, the compound represented by Chemical Formula 1 may include at least one deuterium. That is, at least one R.sub.1 of Chemical Formula 1 may be deuterium, and/or at least one of Ar.sub.1 to Ar.sub.3 may be substituted with one or more deuteriums.

[0055] In Chemical Formula 1, at least one of X.sub.1 to X.sub.3 is N, and preferably all of them are N.

[0056] Preferably, Ar.sub.1 and Ar.sub.2 are each independently a substituted or unsubstituted C.sub.6-20 aryl; or a substituted or unsubstituted C.sub.2-20 heteroaryl containing one or more selected from the group consisting of N, O and S. Each of the Ar.sub.1 and Ar.sub.2 may be substituted with one or more deuteriums.

[0057] Preferably, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenyl; biphenyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl.

[0058] Preferably, R.sub.1 is each independently hydrogen or deuterium.

[0059] Ar.sub.3 is a carbazole substituent represented by Chemical Formula 1-1 or an indolocarbazole substituent represented by Chemical Formula 1-2. The Ar.sub.3 may be substituted with one or more deuteriums.

[0060] When Ar.sub.3 is a substituent represented by Chemical Formula 1-1, preferably, R.sub.11 is each independently hydrogen; deuterium; phenyl; or phenyl substituted with 5 deuteriums.

[0061] When Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p may be an integer of 1 to 8, and R.sub.11 may be deuterium or phenyl.

[0062] Specifically, when Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p may be 1, and R.sub.11 may be phenyl.

[0063] Alternatively, when Ar.sub.3 is a substituent represented by Chemical Formula 1-1, and p is 2 to 8, all of R.sub.11 may be deuterium, or one of R.sub.11 may be phenyl and the rest of R.sub.11 may be deuterium.

[0064] Alternatively, when Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p is an integer of 4 to 8, or 6 to 8, and R.sub.11 may be deuterium.

[0065] Preferably, the indolocarbazole substituent of Chemical Formula 1-2 may be represented by the following Chemical Formula 1-2-a:

##STR00010## [0066] in Chemical Formula 1-2-a, [0067] q, R.sub.12, and Ar.sub.12 are the same as defined in Chemical Formula 1-2.

[0068] When Ar.sub.3 is a substituent represented by Chemical Formula 1-2, R.sub.12 may be each independently hydrogen or deuterium, and Ar.sub.12 may be a substituted or unsubstituted C.sub.6-20 aryl. Preferably, R.sub.12 is each independently hydrogen or deuterium, and Ar.sub.12 is phenyl.

[0069] When Ar.sub.3 is a substituent represented by Chemical Formula 1-2-a, four R.sub.12s may be hydrogen, six R.sub.12s may be deuterium, and Ar.sub.12 may be phenyl.

[0070] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 may be each independently phenyl; phenyl substituted with 5 deuteriums; biphenyl;

biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl.

[0071] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, and Ar.sub.1 and Ar.sub.2 may be each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, and R.sub.1 may be each independently hydrogen or deuterium.

[0072] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p is an integer of 1 to 8, and R.sub.11 may be each independently hydrogen; deuterium; phenyl; or phenyl substituted with 5 deuteriums.

[0073] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p is an integer of 1 to 8, and R.sub.11 may be each independently hydrogen; deuterium; phenyl; or phenyl substituted with 5 deuteriums.

[0074] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p is an integer of 1 to 8, and R.sub.11 may be deuterium or phenyl.

[0075] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 may be a substituent represented by Chemical Formula 1-1, p may be an integer of 6 to 8, and R.sub.11 may be deuterium.

[0076] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-2, q is an integer of 1 to 10, R.sub.12 may be each independently hydrogen or deuterium, and Ar.sub.12 may be phenyl.

[0077] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-2, q is an integer of 1 to 10, R.sub.12 may be each independently hydrogen or deuterium, and Ar.sub.12 may be phenyl.

[0078] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-2-a, q is an integer of 1 to 10, R.sub.12 is each independently hydrogen or deuterium, and Ar.sub.12 may be phenyl.

[0079] In one embodiment, X.sub.1 to X.sub.3 of Chemical Formula 1 are all N, Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenylyl; biphenylyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl, R.sub.1 is each independently hydrogen or deuterium, Ar.sub.3 is a substituent represented by Chemical Formula 1-2-a, q is an integer of 1 to 10, R.sub.12 is each independently hydrogen or deuterium, and Ar.sub.12 may be phenyl.

[0080] Representative examples of the compound represented by Chemical Formula 1 are as follows:

##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016##
##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022##
##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##

##STR00029## ##STR00030## ##STR00031##
##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036## ##STR00037##
##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042## ##STR00043##
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##STR00050## ##STR00051## ##STR00052##
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##STR00065## ##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070##
##STR00071## ##STR00072##
##STR00073## ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##
##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083## ##STR00084##
##STR00085## ##STR00086## ##STR00087## ##STR00088## ##STR00089## ##STR00090##
##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096##
##STR00097## ##STR00098## ##STR00099##
##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105##
##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111##
##STR00112## ##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117##
##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123##
##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##
##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135##
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##STR00157## ##STR00158## ##STR00159## ##STR00160##
##STR00161## ##STR00162## ##STR00163## ##STR00164## ##STR00165## ##STR00166##
##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171## ##STR00172##
##STR00173## ##STR00174## ##STR00175## ##STR00176##

[0081] The compound represented by Chemical Formula 1 can be prepared by a preparation method as shown in the following Reaction Scheme 1 as an example.

##STR00177##

[0082] In Reaction Scheme 1, the remaining substituents except for X' are the same as defined above, and X' is halogen, preferably chloro or bromo.

[0083] A first step of Reaction Scheme 1 is a Suzuki coupling reaction, which is preferably carried out in the presence of a palladium catalyst and a base, and a reactive group for the Suzuki coupling reaction can be modified as known in the art.

[0084] A second step of Reaction Scheme 1 is an amine substitution reaction, which is preferably carried out in the presence of a palladium catalyst and a base, and a reactive group for the amine substitution reaction can be modified as known in the art.

[0085] When the compound represented by Chemical Formula 1 contains deuterium, it can be prepared from a precursor containing deuterium, or it can be prepared from a precursor containing no deuterium, and then finally contain deuterium through a hydrogen/deuterium exchange reaction.

[0086] The preparation method of the compound represented by Chemical Formula 1 can be further embodied in Preparation Examples described hereinafter.

[0087] The compound represented by Chemical Formula 2 is characterized in that it has at least one deuterium.

[0088] That is, in Chemical Formula 2, at least one of R'.sub.1 and R'.sub.2 is deuterium; or at least one of R'.sub.1 and R'.sub.2 is deuterium, and at least one substituent of Ar'.sub.1 and Ar'.sub.2 includes one or more deuteriums; or when both R'.sub.1 and R'.sub.2 are not deuterium, at least one substituent of Ar'.sub.1 and Ar'.sub.2 may include one or more deuteriums.

[0089] Preferably, R'.sub.1 and R'.sub.2 are each independently hydrogen; deuterium; or a substituted or unsubstituted C.sub.6-12 aryl. Wherein, the aryl may be substituted with one or more deuteriums.

[0090] Preferably, R'.sub.1 and R'.sub.2 are each independently hydrogen; deuterium; or phenyl unsubstituted or substituted with 1 to 5 deuteriums.

[0091] Preferably, at least one of R'.sub.1 and R'.sub.2 is deuterium.

[0092] Preferably, n' and m' are each independently an integer of 5 to 7, or 6 to 7, and both R'.sub.1 and R'.sub.2 are deuterium.

[0093] Preferably, Ar'.sub.1 and Ar'.sub.2 are each independently phenyl unsubstituted or substituted with 1 to 5 deuteriums; biphenyl unsubstituted or substituted with 1 to 9 deuteriums; naphthyl unsubstituted or substituted with 1 to 7 deuteriums; dimethylfluorenyl unsubstituted or substituted with 1 to 13 deuteriums; dibenzofuranyl unsubstituted or substituted with 1 to 7 deuterium; or dibenzothiophenyl unsubstituted or substituted with 1 to 7 deuteriums.

[0094] In one embodiment, n' and m' are each independently an integer of 6 to 7, both R'.sub.1 and R'.sub.2 are deuterium, and Ar'.sub.1 and Ar'.sub.2 are each independently phenyl unsubstituted or substituted with 1 to 5 deuterium; biphenyl unsubstituted or substituted with 1 to 9 deuteriums; naphthyl unsubstituted or substituted with 1 to 7 deuteriums; dimethylfluorenyl unsubstituted or substituted with 1 to 13 deuteriums; dibenzofuranyl unsubstituted or substituted with 1 to 7 deuteriums; or dibenzothiophenyl unsubstituted or substituted with 1 to 7 deuteriums.

[0095] Preferably, the deuterium substitution rate of Chemical Formula 2 is 40 to 100%. The deuterium substitution rate means the number of substituted deuteriums contained in Chemical Formula 2 relative to the total number of hydrogens that can be present in Chemical Formula 2. Preferably, the deuterium substitution rate of Chemical Formula 2 is 60% or more, 65% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, and 99% or less, 98% or less, 97% or less, 96% or less, 95% or less, 94% or less, 93% or less, or 92% or less,

[0096] Representative examples of the compound represented by Chemical Formula 2 are as follows:

##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183##
##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188## ##STR00189##
##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195##
##STR00196## ##STR00197## ##STR00198## ##STR00199## ##STR00200## ##STR00201##
##STR00202## ##STR00203## ##STR00204## ##STR00205## ##STR00206## ##STR00207##
##STR00208## ##STR00209## ##STR00210##

[0097] wherein, methyl groups contained in each of the above listed compounds including the same are each independently CH.sub.3, CH.sub.2D, CHD.sub.2, or CD.sub.3. As an example, in the compound of

##STR00211##

the two methyl groups of the dimethylfluorenyl group may be each independently CH.sub.3, CH.sub.2D, CHD.sub.2, or CD.sub.3.

[0098] The compound represented by Chemical Formula 2 can be prepared by a preparation method as shown in the following Reaction Scheme 2 as an example.

##STR00212##

[0099] In Reaction Scheme 2, the remaining substituents except for X'' are the same as defined above, and X'' is halogen, preferably bromo or chloro.

[0100] Reaction Scheme 2 is a Suzuki coupling reaction, which is preferably carried out in the presence of a palladium catalyst and a base, and a reactive group for the Suzuki coupling reaction can be modified as known in the art.

[0101] The compound represented by Chemical Formula 2 can be prepared from a precursor containing deuterium, or can be prepared from a precursor containing no deuterium, and finally contain deuterium through a hydrogen/deuterium exchange reaction.

[0102] The preparation method of the compound represented by Chemical Formula 2 can be further embodied in Preparation Examples described hereinafter.

[0103] In the light emitting layer, the weight ratio between the compound represented by Chemical Formula 1 to the compound represented by Chemical Formula 2 is 1:99 to 99:1, 5:95 to 95:5, or 10:90 to 90:10.

[0104] Meanwhile, the compound represented by Chemical Formula 1 and the compound represented by Chemical Formula 2 may be included in the light emitting layer as a simple mixture or may be included in the light emitting layer as an organic alloy. The organic alloy is a material obtained by pre-treating more than two single organic compounds, and may have a chemical interaction between the single organic compounds due to the pre-treatment. The pre-treatment may be a heat treatment such as heating and sublimation followed by cooling, but is not limited thereto.

[0105] The organic alloy has a chemical interaction between two or more single organic compounds as described above and thus, has different intrinsic characteristics from the single organic compounds and their simple mixture having no chemical interaction between single organic compounds. Herein, the simple mixture means a mixture obtained by physically mixing single organic compounds without any pre-treatment. In other words, a simple mixture of the first organic compound and the second organic compound exhibits the characteristics of the first organic compound, the second organic compound, or a combination thereof, while the organic alloy of the first organic compound and the second organic compound can exhibit different characteristics from those of the first organic compound, the second organic compound, or a simple mixture thereof.

[0106] In one example, the emission wavelength of the organic alloy may differ from the emission wavelength of the first organic compound, the second organic compound and the simple mixture thereof.

[0107] Further, the color of the organic alloy may be different from the color of the first organic compound, the second organic compound, and the simple mixture thereof.

[0108] Furthermore, the glass transition temperature (T_g) of the organic alloy may be different from the glass transition temperature (T_g) of the first organic compound, the second organic compound, and the simple mixture thereof. In addition, the organic alloy may have a different crystallization temperature (T_c) from that of the first organic compound, the second organic compound, and the simple mixture thereof. Further, the organic alloy may have a different melting point (T_m) from that of the first organic compound, the second organic compound, and the simple mixture thereof.

[0109] The organic alloy can be pre-treated in various methods, and for example, can be obtained from the steps of heat-treating the first organic compound and the second organic compound and liquidating or gasifying the first and second organic compounds and then, cooling and solidifying the heat-treated compounds. Also, an additional step of physically grounding the organic alloy obtained as a solid like a mass using a mixer and the like can be included.

[0110] The organic alloy is a product obtained through the aforementioned pre-treatment and may be supplied by using one source to form a thin film. Accordingly, this eliminates the need for a process control step required when two or more organic compounds are respectively supplied from separate sources, and thus, the process can be simplified.

[0111] In addition, the organic alloy is a product obtained through the aforementioned pre-treatment and thus, can secure uniformity and consistency for deposition as compared with the two or more single organic compounds supplied by using two or more separate sources or their simple mixture of supplied by using one single source. Accordingly, when a plurality of thin films are formed through a continuous process, the organic alloy may be used to continuously produce the thin films including components in a substantially equivalent ratio and thus, increase reproducibility and reliability of the thin films.

[0112] The dopant material is not particularly limited as long as it is a material used for the organic light emitting device. As an example, an aromatic amine derivative, a styrylamine compound, a

boron complex, a fluoranthene compound, a metal complex, and the like can be mentioned. Specific examples of the aromatic amine derivatives include substituted or unsubstituted fused aromatic ring derivatives having an arylamino group, examples thereof include pyrene, anthracene, chrysene, and perflanthene having the arylamino group, and the like. The styrylamine compound is a compound of an arylamine, which is unsubstituted or substituted with one or two or more substituent groups selected from the group consisting of an aryl group, a silyl group, an alkyl group, a cycloalkyl group, and an arylamino group are substituted or unsubstituted, substituted with at least one arylvinyl group. Specific examples thereof include styrylamine, styryldiamine, styryltriamine, styryltetramine, and the like, but are not limited thereto. Further, examples of the metal complex include an iridium complex, a platinum complex, and the like, but are not limited thereto.

Hole Transport Layer

[0113] The organic light emitting device according to the present disclosure may further include a hole transport layer between the light emitting layer and the anode.

[0114] The hole transport layer is a layer that receives holes from a hole injection layer and transports the holes to the light emitting layer. The hole transport material is suitably a material having large mobility to the holes, which may receive holes from the anode or the hole injection layer and transfer the holes to the light emitting layer.

[0115] Specific examples of the hole transport material include an arylamine-based organic material, a conductive polymer, a block copolymer in which a conjugate portion and a non-conjugate portion are present together, and the like, but are not limited thereto.

Hole Injection Layer

[0116] The organic light emitting device according to the present disclosure may further include a hole injection layer between the anode and the hole transport layer, if necessary.

[0117] The hole injection layer is a layer injecting holes from an electrode, and the hole injection material is preferably a compound which has a capability of transporting the holes, has a hole injection effect in the anode and an excellent hole injection effect to the light emitting layer or the light emitting material, prevents movement of an exciton generated in the light emitting layer to the electron injection layer or the electron injection material, and is excellent in the ability to form a thin film. Further, it is preferable that a HOMO (highest occupied molecular orbital) of the hole injection material is between the work function of the anode material and a HOMO of a peripheral organic material layer.

[0118] Specific examples of the hole injection material include metal porphyrine, oligothiophene, an arylamine-based organic material, a hexanitrihexaazatriphenylene-based organic material, a quinacridone-based organic material, a perylene-based organic material, anthraquinone, polyaniline and polythiophene-based conductive polymer, and the like, but are not limited thereto.

Electron Blocking Layer

[0119] The organic light emitting device according to the present disclosure includes an electron blocking layer between the hole transport layer and the light emitting layer, if necessary.

[0120] The electron blocking layer prevents the electrons injected from the cathode from being transferred to the hole transport layer without being recombined in the light emitting layer, which may also be referred to as an electron inhibition layer. The electron blocking layer is preferably a material having the smaller electron affinity than the electron transport layer.

Electron Transport Layer

[0121] The organic light emitting device according to the present disclosure may include an electron transport layer between the light emitting layer and the cathode, if necessary.

[0122] The electron transport layer is a layer that receives the electrons from the cathode or the electron injection layer formed on the cathode and transports the electrons to the light emitting layer, and that suppress the transfer of holes from the light emitting layer, and an electron transport material is suitably a material which may receive electrons well from a cathode and transfer the

electrons to a light emitting layer, and has a large mobility for electrons.

[0123] Specific examples of the electron transport material include: an Al complex of 8-hydroxyquinoline: a complex including Alq.sub.3: an organic radical compound: a hydroxyflavone-metal complex, and the like, but are not limited thereto. The electron transport layer may be used with any desired cathode material, as used according to a conventional technique. In particular, appropriate examples of the cathode material are a typical material which has a low work function, followed by an aluminum layer or a silver layer. Specific examples thereof include cesium, barium, calcium, ytterbium, and samarium, in each case followed by an aluminum layer or a silver layer.

Electron Injection Layer

[0124] The organic light emitting device according to the present disclosure may further include an electron injection layer between the electron transport layer and the cathode.

[0125] The electron injection layer is a layer which injects electrons from an electrode, and is preferably a compound which has a capability of transporting electrons, has an effect of injecting electrons from a cathode and an excellent effect of injecting electrons into a light emitting layer or a light emitting material, prevents excitons produced from the light emitting layer from moving to a hole injection layer, and is also excellent in the ability to form a thin film.

[0126] Specific examples of the electron injection layer include fluorenone, anthraquinodimethane, diphenoquinone, thiopyran dioxide, oxazole, oxadiazole, triazole, imidazole, perylenetetracarboxylic acid, fluorenylidene methane, anthrone, and the like, and derivatives thereof, a metal complex compound, a nitrogen-containing 5-membered ring derivative, and the like, but are not limited thereto.

[0127] Examples of the metal complex compound include 8-hydroxyquinolinato lithium, bis(8-hydroxy quinolinato)zinc, bis(8-hydroxy quinolinato)copper, bis(8-hydroxyquinolinato)manganese, tris(8-hydroxyquinolinato)aluminum, tris(2-methyl-8-hydroxyquinolinato)aluminum, tris(8-hydroxy quinolinato)gallium, bis(10-hydroxybenzo[h]quinolinato)beryllium, bis(10-hydroxybenzo[h]quinolinato)zinc, bis(2-methyl-8-quinolinato)chlorogallium, bis(2-methyl-8-quinolinato)(o-cresolato)gallium, bis(2-methyl-8-quinolinato)(1-naphtholato)aluminum, bis(2-methyl-8-quinolinato)(2-naphtholato)gallium, and the like, but are not limited thereto.

Hole Blocking Layer

[0128] The organic light emitting device according to the present disclosure may include a hole blocking layer between the electron transport layer and the light emitting layer, if necessary.

[0129] The hole blocking layer prevents the holes injected from the anode from being transferred to the electron transport layer without being recombined in the light emitting layer, and the hole blocking layer is preferably a material having high ionization energy.

Organic Light Emitting Device

[0130] The structure of the organic light emitting device according to the present disclosure is illustrated in FIG. 1. FIG. 1 shows an example of an organic light emitting device comprising a substrate 1, an anode 2, a light emitting layer 3, and a cathode 4. Further, FIG. 2 shows an example of an organic light emitting device comprising a substrate 1, an anode 2, a hole injection layer 5, a hole transport layer 6, a light emitting layer 3, an electron transport layer 7, an electron injection layer 8 and a cathode 4. Further, FIG. 3 shows an example of an organic light emitting device comprising a substrate 1, an anode 2, a hole injection layer 5, a hole transport layer 6, an electron blocking layer 9, a light emitting layer 3, a hole blocking layer 10, an electron transport layer 7, an electron injection layer 8 and a cathode 4.

[0131] The organic light emitting device according to the present disclosure can be manufactured by sequentially stacking the above-described structures. In this case, the organic light emitting device may be manufactured by depositing a metal, metal oxides having conductivity, or an alloy thereof on the substrate by using a PVD (physical vapor deposition) method such as a sputtering method or an e-beam evaporation method to form the anode, forming the respective layers described above thereon, and then depositing a material that can be used as the cathode thereon.

[0132] In addition to such a method, the organic light emitting device may be manufactured by sequentially depositing from the cathode material to the anode material on a substrate in the reverse order of the above-mentioned configuration (WO 2003/012890). Further, the light emitting layer may be formed by subjecting hosts and dopants to a vacuum deposition method and a solution coating method. Herein, the solution coating method means a spin coating, a dip coating, a doctor blading, an inkjet printing, a screen printing, a spray method, a roll coating, or the like, but is not limited thereto.

[0133] Meanwhile, the organic light emitting device according to the present disclosure may be a bottom emission device, a top emission device, or a double-sided light emitting device, and particularly, may be a bottom emission device that requires relatively high luminous efficiency.

[0134] The preparation of the organic light emitting device according to the present disclosure will be described in detail in the following examples. However, these examples are presented for illustrative purposes only, and are not intended to limit the scope of the present disclosure.

PREPARATION EXAMPLE 1: Preparation of the Compound of Chemical Formula 1

Preparation Example 1-1: Synthesis of Compound 1-1

Step 1) Synthesis of Intermediate 1-1-1

##STR00213##

[0135] 6-Bromo-3-chlorodibenzo[b,d]furan (20 g, 71 mmol) and bis(pinacolato)diboron (18 g, 71 mmol) were added to 400 ml of dioxane under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, anhydrous potassium acetate (20.9 g, 213.1 mmol) was added thereto, sufficiently stirred, and then palladium dibenzylideneacetone palladium (0.8 g, 1.4 mmol) and tricyclohexylphosphine (0.8 g, 2.8 mmol) were added. After the reaction for 7 hours, the reaction mixture was cooled to room temperature and the organic layer was filtered to remove a salt, and the filtered organic layer was distilled. This was again added to and dissolved in 233 mL of chloroform, and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by a silica gel column chromatography to give Intermediate 1-1-1 (15.2 g, 65%, MS: [M+H].sup.+ = 329.6).

Step 2) Synthesis of Intermediate 1-1-2

##STR00214##

[0136] 1-1-1 (20 g, 60.9 mmol) and 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine (23.1 g, 60.9 mmol) were added to tetrahydrofuran (400 ml) under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (25.2 g, 182.6 mmol) was dissolved in water (75 ml), added thereto, sufficiently stirred, and then tetrakis(triphenyl)phosphinopalladium (2.1 g, 1.8 mmol) was added. After reaction for 2 hours, the mixture was cooled to room temperature, and then the resulting solid was filtered. The solid was added to 664 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was recrystallized from chloroform and ethyl acetate to give a white solid compound 1-1-2 (24.5 g, yield: 74%; MS: [M+H].sup.+ = 546.1).

Step 3) Synthesis of Compound 1-1

##STR00215##

[0137] 1-1-2 (15 g, 27.5 mmol) and 9H-carbazole (4.6 g, 27.5 mmol) were added to 300 ml of toluene under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, sodium tert-butoxide (7.9 g, 82.6 mmol) was added thereto and sufficiently stirred, and then bis(tri-tert-butylphosphine) palladium (0.4 g, 0.8 mmol) was added. After the reaction for 2 hours, the reaction mixture was cooled to room temperature and then the organic layer was subjected to filtration treatment to remove a salt, and then the filtered organic layer was distilled. This was added again to 185 mL of chloroform, dissolved and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was

distilled under reduced pressure. The concentrated compound was purified by silica gel column and then subjected to sublimation purification to give Compound 1-1 (10 g, 54%, MS: [M+H].sup.=671.8).

Preparation Example 1-2: Synthesis of Compound 1-2

Step 1) Synthesis of Intermediate 1-2-1

##STR00216##

[0138] 6-Bromo-3-chlorodibenzo[b,d]furan (15 g, 53.3 mmol) and 9H-carbazole-d8 (9.3 g, 53.3 mmol) were added to 300 ml of toluene under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, sodium tert-butoxide (15.4 g, 159.8 mmol) was added thereto and sufficiently stirred, and then bis(tri-tert-butylphosphine) palladium (0.8 g, 1.6 mmol) was added. After the reaction for 3 hours, the reaction mixture was cooled to room temperature and then the organic layer was subjected to filtration treatment to remove a salt, and then the filtered organic layer was distilled. This was added again to 200 mL of chloroform, dissolved and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by silica gel column to give Compound 1-2-1 (10.8 g, yield: 54%, MS: [M+H].sup.=376.9).

Step 2) Synthesis of Intermediate 1-2-2

##STR00217##

[0139] 1-2-1 (20 g, 53.2 mmol) and bis(pinacolato)diboron (13.5 g, 53.2 mmol) were added to 400 ml of dioxane under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, anhydrous potassium acetate (15.7 g, 159.6 mmol) was added thereto, sufficiently stirred, and then palladium dibenzylideneacetone palladium (0.6 g, 1.1 mmol) and tricyclohexylphosphine (0.6 g, 2.1 mmol) were added. After the reaction for 3 hours, the reaction mixture was cooled to room temperature and the organic layer was subjected to filtration treatment to remove a salt, and the filtered organic layer was distilled. This was again added to 249 mL of chloroform, dissolved, and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by a silica gel column chromatography to give Intermediate 1-2-2 (18.4 g, yield: 74%, MS: [M+H].sup.=468.4).

Step 3) Synthesis of Compound 1-2

##STR00218##

[0140] 1-2-2 (20 g, 42.8 mmol) and 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine (14.7 g, 42.8 mmol) were added to 400 ml of tetrahydrofuran under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (17.7 g, 128.4 mmol) was dissolved in 54 ml of water, added thereto, sufficiently stirred, and then tetrakis(triphenyl)phosphinepalladium (1.5 g, 1.3 mmol) was added. After the reaction for 3 hours, the reaction mixture was cooled to room temperature and then the resulting solid was filtered. The solid was added to 555 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by silica gel column chromatography and then subjected to sublimation purification to give Compound 1-2 (21.4 g, yield: 77%; MS: [M+H].sup.=649.8).

Preparation Example 1-3: Synthesis of Compound 1-3

##STR00219##

[0141] Compound 1-3 (MS: [M+H].sup.=669.8) was prepared in the same manner as in the preparation method of compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 1-bromo-7-chlorodibenzo[b,d]furan-2,3,4,6,8,9-d6, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d5)-1,3,5-triazine was changed to 2-chloro-4-(dibenzo[b,d]furan-3-yl)-6-phenyl-1,3,5-triazine, and 9H-carbazole was changed to 9H-carbazole-d8.

Preparation Example 1-4: Synthesis of Compound 1-4

##STR00220##

[0142] Compound 1-4 (MS: [M+H]⁺=661.8) was prepared in the same manner as in the preparation method of Compound 1-2, except that in Preparation Example 1-2, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 1-bromo-7-chlorodibenzo[b,d]furan-2,3,4,6,8,9-d₆, 9H-carbazole-d₈ was changed to 9H-carbazole, and 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine was changed to 2-chloro-4-(dibenzo[b,d]furan-1-yl)-6-phenyl-1,3,5-triazine.

Preparation Example 1-5: Synthesis of Compound 1-5

##STR00221##

[0143] Compound 1-5 (MS: [M+H]⁺=646.8) was prepared in the same manner as in the preparation method of Compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 2-bromo-6-chlorodibenzo[b,d]furan, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine was changed to 2-chloro-4,6-diphenyl-1,3,5-triazine, and 9H-carbazole was changed to 2-(phenyl-d₅)-9H-carbazole.

Preparation Example 1-6: Synthesis of Compound 1-6

##STR00222##

[0144] Compound 1-6 (MS: [M+H]⁺=573.7) was prepared in the same manner as in the preparation method of Compound 1-2, except that in Preparation Example 1-2, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 2-bromo-6-chlorodibenzo[b,d]furan, and 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine was changed to 2-chloro-4,6-diphenyl-1,3,5-triazine.

Preparation Example 1-7: Synthesis of Compound 1-7

##STR00223##

[0145] Compound 1-7 (MS: [M+H]⁺=648.8) was prepared in the same manner as in the preparation method of Compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 1-bromo-6-chlorodibenzo[b,d]furan, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine was changed to 2-chloro-4,6-diphenyl-1,3,5-triazine, and 9H-carbazole was changed to 4-phenyl-9H-carbazole-1,2,3,5,6,7,8-d₇.

Preparation Example 1-8: Synthesis of Compound 1-8

##STR00224##

[0146] Compound 1-8 (MS: [M+H]⁺=646.8) was prepared in the same manner as in the preparation method of Compound 1-2, except that in Preparation Example 1-2, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 1-bromo-6-chlorodibenzo[b,d]furan, 9H-carbazole-d₈ was changed to 9H-carbazole, and 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine was changed to 2-([1,1'-biphenyl]-4-yl)-4-chloro-6-(phenyl-d₅)-1,3,5-triazine.

Preparation Example 1-9: Synthesis of Compound 1-9

##STR00225##

[0147] Compound 1-9 (MS: [M+H]⁺=651.8) was prepared in the same manner as in the preparation method of Compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 8-bromo-1-chlorodibenzo[b,d]furan, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine was changed to 2-chloro-4,6-bis(phenyl-d₅)-1,3,5-triazine, and 9H-carbazole was changed to 3-phenyl-9H-carbazole.

Preparation Example 1-10: Synthesis of Compound 1-10

##STR00226##

[0148] Compound 1-10 (MS: [M+H]⁺=573.7) was prepared in the same manner as in the preparation method of Compound 1-2, except that in Preparation Example 1-2, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 2-bromo-7-chlorodibenzo[b,d]furan, and 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine was changed to 2-chloro-4,6-diphenyl-1,3,5-triazine.

Preparation Example 1-11: Synthesis of Compound 1-11

##STR00227##

[0149] Compound 1-11 (MS: [M+H]⁺=746.0) was prepared in the same manner as in the preparation method of Compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 4-bromo-6-chlorodibenzo[b,d]thiophene, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine was changed to 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-(phenyl-d₅)-1,3,5-triazine, and 9H-carbazole was changed to 4-phenyl-9H-carbazole-1,2,3,5,6,7,8-d₇.

Preparation Example 1-12: Synthesis of Compound 1-12

##STR00228##

[0150] Compound 1-12 (MS: [M+H]⁺=665.9) was prepared in the same manner as in the preparation method of Compound 1-1, except that in Preparation Example 1-1, 6-bromo-3-chlorodibenzo[b,d]furan was changed to 6-bromo-1-chlorodibenzo[b,d]thiophene, 2-chloro-4-(dibenzo[b,d]thiophen-4-yl)-6-(phenyl-d₅)-1,3,5-triazine was changed to 2-([1,1'-biphenyl]-3-yl)-4-chloro-6-phenyl-1,3,5-triazine, and 9H-carbazole was changed to 9H-carbazole-1,2,3,4,5,6,7,8-d₈.

PREPARATION EXAMPLE 2: Preparation of the Compound of Chemical Formula 2

Preparation Example 2-1: Synthesis of Compound 2-1

Step 1) Synthesis of Intermediate 2-1-1

##STR00229##

[0151] 9-([1,1'-Biphenyl]-4-yl)-3-bromo-9H-carbazole (20 g, 50.4 mmol) and (9-phenyl-9H-carbazol-3-yl) boronic acid (14.5 g, 50.4 mmol) were added to 400 ml of tetrahydrofuran under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (20.9 g, 151.1 mmol) was dissolved in 63 ml of water, added thereto and sufficiently stirred, and then tetrakis(triphenyl)-phosphinopalladium (1.7 g, 1.5 mmol) was added. After the reaction for 5 hours, the reaction mixture was cooled to room temperature, and the resulting solid was filtered. The solid was added to 565 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was recrystallized from chloroform and ethyl acetate to give a white solid compound 2-1-1 (19.2 g, yield: 68%; MS: [M+H]⁺=561.7).

Step 2) Synthesis of Compound 2-1

##STR00230##

[0152] Compound 2-1-1 (10.0 g, 17.8 mmol), PtO.sub.2 (1.2 g, 5.4 mmol) and D.sub.2O (89 ml) were placed in a shaker tube, and the tube was sealed and heated at 250° C. and 600 psi for 12 hours. When the reaction was completed, chloroform was added thereto, and the reaction solution was transferred to a separatory funnel and extracted. The extract was dried over anhydrous magnesium sulfate, concentrated, and the sample was purified by silica gel column chromatography and then subjected to sublimation purification to give 3.9 g of Compound 2-1. (yield: 38%, MS [M+H]⁺=579.8)

Preparation Example 2-2: Synthesis of Compound 2-2

Step 1) Synthesis of Intermediate 2-2-1

##STR00231##

[0153] Compound 9-([1,1'-biphenyl]-4-yl)-3-bromo-9H-carbazole (20 g, 50.4 mmol) and 200 ml of Benzen-D₆ were added under a nitrogen atmosphere, and the mixture was stirred. Then, triflic acid (3.8 g, 25.1 mmol) was added, allowed to warm and then stirred. After the reaction for 4 hours, the reaction mixture was cooled to room temperature, and the resulting solid was filtered. The solid was added to 886 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by silica column to give a white solid compound 2-2-1 (11.3 g, yield: 55%; MS: [M+H]⁺=409.4).

Step 2) Synthesis of Compound 2-2

##STR00232##

[0154] 2-2-1 (20 g, 49 mmol) and (9-([1,1'-biphenyl]-2-yl)-9H-carbazol-3-yl) boronic acid (17.8 g, 49 mmol) were added to 400 ml of tetrahydrofuran under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (20.3 g, 146.9 mmol) was dissolved in 60 ml of water, added thereto, sufficiently stirred, and then tetrakis(triphenyl)-phosphinopalladium (1.7 g, 1.5 mmol) was added. After reaction for 2 hours, the mixture was cooled to room temperature, and then the resulting solid was filtered. The solid was added to 634 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was purified by silica gel column chromatography and then subjected to sublimation purification to give Compound 2-2 (24.7 g, yield: 78%, MS [M+H].sup.+ = 647.9)

Preparation Example 2-3: Synthesis of Compound 2-3

Step 1) Synthesis of Intermediate 2-3-1

##STR00233##

[0155] 9-([1,1'-biphenyl]-3-yl)-3-bromo-9H-carbazole (20 g, 50.2 mmol) and (9-([1,1'-biphenyl]-3-yl)-9H-carbazol-3-yl) boronic acid (18.2 g, 50.2 mmol) were added to 400 ml of tetrahydrofuran under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (20.8 g, 150.6 mmol) was dissolved in 21 ml of water, added thereto, and sufficiently stirred, and then tetrakis(triphenyl)-phosphinopalladium (1.7 g, 1.5 mmol) was added. After reaction for 5 hours, the mixture was cooled to room temperature, and then the resulting solid was filtered. The solid was added to 640 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was recrystallized from chloroform and ethyl acetate to give a solid compound 2-3-1 (24.3 g, yield: 76%, MS: [M+H].sup.+ = 637.8).

Step 2) Synthesis of Compound 2-3

##STR00234##

[0156] 2-3-1 (10.0 g, 15.7 mmol), PtO.sub.2 (1.1 g, 4.8 mmol) and D.sub.2O (89 ml) were placed in a shaker tube, and the tube was sealed and heated at 250° C. and 600 psi for 12 hours. When the reaction was completed, chloroform was added thereto, and the reaction solution was transferred to a separatory funnel and extracted. The extract was dried over anhydrous magnesium sulfate, concentrated, and the sample was purified by silica gel column chromatography and then subjected to sublimation purification to give Compound 2-3 (5.7 g, yield: 55%, MS [M+H].sup.+ = 657.9)

Preparation Example 2-4: Synthesis of Compound 2-4

Step 1) Synthesis of Intermediate 2-4-1

##STR00235##

[0157] 3-Bromo-9H-carbazole (20 g, 81.3 mmol) and (9-phenyl-9H-carbazol-3-yl) boronic acid (23.3 g, 81.3 mmol) were added to 400 ml of tetrahydrofuran under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, potassium carbonate (33.7 g, 243.8 mmol) was dissolved in 102 ml of water, added thereto, and sufficiently stirred, and then tetrakis(triphenyl)-phosphinopalladium (2.8 g, 2.4 mmol) was added. After reaction for 2 hours, the mixture was cooled to room temperature, and then the resulting solid was filtered. The solid was added to 664 mL of chloroform, dissolved, washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated compound was recrystallized from chloroform and ethyl acetate to give a white solid compound 2-4-1 (21.6 g, yield: 65%, MS: [M+H].sup.+ = 409.5).

Step 2) Synthesis of Compound 2-4-2

##STR00236##

[0158] Compound 2-4-1 (10.0 g, 24.5 mmol), PtO.sub.2 (1.7 g, 7.5 mmol) and D.sub.2O (139 ml) were placed in a shaker tube, and the tube was sealed and heated at 250° C. and 600 psi for 12

hours. When the reaction was completed, chloroform was added thereto, and the reaction solution was transferred to a separatory funnel and extracted. The extract was dried over anhydrous magnesium sulfate, concentrated, and the sample was purified by silica gel column chromatography and then subjected to sublimation purification to give 5.2 g of Compound 2-4-2 (yield: 50%, MS [M+H].sup.+ = 422.6)

Step 3) Synthesis of Compound 2-4

##STR00237##

[0159] 2-4-2 (15 g, 35.6 mmol) and 2-chloro-9,9-dimethyl-9H-fluorene (8.1 g, 35.6 mmol) were added to 300 ml of toluene under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, sodium tert-butoxide (10.3 g, 106.7 mmol) was added thereto and sufficiently stirred, and then bis(tri-tert-butylphosphine) palladium (0.5 g, 1.1 mmol) was added. After the reaction for 1 hour, the reaction mixture was cooled to room temperature and then the organic layer was subjected to filtration treatment to remove a salt, and then the filtered organic layer was distilled. This was added again to 218 mL of chloroform, dissolved and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated sample was purified by silica gel column and then subjected to sublimation purification to give Compound 2-4 (15.5 g, 71%, MS: [M+H]⁺ = 614.8).

Preparation Example 2-5: Synthesis of Compound 2-5

Step 1) Synthesis of Compound 2-5

##STR00238##

[0160] 2-4-2 (15 g, 35.6 mmol) and 2-bromodibenzo[b,d]furan (8.8 g, 35.6 mmol) were added to 300 ml of toluene under a nitrogen atmosphere, and the mixture was stirred and refluxed. Then, sodium tert-butoxide (10.3 g, 106.7 mmol) was added thereto and sufficiently stirred, and then bis(tri-tert-butylphosphine) palladium (0.5 g, 1.1 mmol) was added. After the reaction for 3 hours, the reaction mixture was cooled to room temperature and then the organic layer was subjected to filtration treatment to remove a salt, and then the filtered organic layer was distilled. This was added again to 209 mL of chloroform, dissolved and washed twice with water. The organic layer was then separated, anhydrous magnesium sulfate was added, stirred and then filtered. The filtrate was distilled under reduced pressure. The concentrated sample was purified by silica gel column and then subjected to sublimation purification to give Compound 2-5 (16.5 g, 79%, MS: [M+H]⁺ = 588.8).

PREPARATION EXAMPLE 3: Preparation of Organic Alloy

Preparation Example 3-1: Preparation of Organic Alloy 1

[0161] Compound 1-1 and Compound 2-1 were mixed and placed in a weight ratio of 40:60 in a vacuum chamber. The two mixtures were melted by increasing the temperature under a pressure of 10⁻² Torr or less, and then the mixture was cooled to room temperature after 1 hour to obtain a solid product. This product was ground with a mixer to obtain a powdery organic alloy 1.

Preparation Examples 3-2 to 3-18 and Preparation Examples 3-A to 3-D

[0162] Organic alloys 2 to 18 and organic alloys A to D were prepared in the same manner as in the preparation method of organic alloy 1, except that the materials to be mixed were changed as shown in Table 1 below. Compounds H-1, H-2 and H-3 of Table 1 are as follows.

##STR00239##

TABLE-US-00001

TABLE 1	Mixing ratio	Preparation Organic	Mixed	Mixed (weight	Example
alloy material	1 material	2 material	ratio)	Preparation Organic	Compound 1-1
Example 3-1 alloy 1	Preparation Organic	Compound 1-2	Compound 2-1	40:60	
Example 3-2 alloy 2	Preparation Organic	Compound 1-2	Compound 2-2	40:60	
Example 3-3 alloy 3	Preparation Organic	Compound 1-2	Compound 2-3	40:60	
Example 3-4 alloy 4	Preparation Organic	Compound 1-3	Compound 2-2	40:60	
Example 3-5 alloy 5	Preparation Organic	Compound 1-4	Compound 2-1	40:60	
Example 3-6 alloy 6	Preparation Organic	Compound 1-4	Compound 2-2	40:60	
Example 3-7					

alloy 7 Preparation Organic Compound 1-5 Compound 2-4 40:60 Example 3-8 alloy 8 Preparation Organic Compound 1-6 Compound 2-5 40:60 Example 3-9 alloy 9 Preparation Organic Compound 1-7 Compound 2-1 40:60 Example 3-10 alloy 10 Preparation Organic Compound 1-7 Compound 2-2 40:60 Example 3-11 alloy 11 Preparation Organic Compound 1-8 Compound 2-5 40:60 Example 3-12 alloy 12 Preparation Organic Compound 1-9 Compound 2-4 40:60 Example 3-13 alloy 13 Preparation Organic Compound 1-10 Compound 2-1 40:60 Example 3-14 alloy 14 Preparation Organic Compound 1-10 Compound 2-2 40:60 Example 3-15 alloy 15 Preparation Organic Compound 1-11 Compound 2-2 40:60 Example 3-16 alloy 16 Preparation Organic Compound 1-11 Compound 2-3 40:60 Example 3-17 alloy 17 Preparation Organic Compound 1-12 Compound 2-3 40:60 Example 3-18 alloy 18 Preparation Organic Compound H-1 Compound H-2 40:60 Example 3-A alloy A Preparation Organic Compound H-1 Compound H-3 40:60 Example 3-B alloy B Preparation Organic Compound 1-1 Compound H-2 40:60 Example 3-C alloy C Preparation Organic Compound 1-1 Compound H-3 40:60 Example 3-D alloy D

EXAMPLES

Example 1: Manufacture of Organic Light Emitting Device

[0163] A glass substrate on which a thin film of ITO (indium tin oxide) was coated in a thickness of 1400 Å was put into distilled water containing the detergent dissolved therein and washed by the ultrasonic wave. In this case, the used detergent was a product commercially available from Fischer Co. and the distilled water was one which had been twice filtered by using a filter commercially available from Millipore Co. The ITO was washed for 30 minutes, and ultrasonic washing was then repeated twice for 10 minutes by using distilled water. After the washing with distilled water was completed, the substrate was ultrasonically washed with isopropyl alcohol, acetone, and methanol solvent, and dried, after which it was transported to a plasma cleaner. Then, the substrate was cleaned with oxygen plasma for 5 minutes, and then transferred to a vacuum evaporator.

[0164] On the ITO transparent electrode thus prepared, 95 wt % of HT-A and 5 wt % of PD were thermally vacuum-deposited to a thickness of 100 Å to form a hole injection layer, and then only HT-A material was deposited to a thickness of 1150 Å to form a hole transport layer. The following HT-B was thermally vacuum-deposited to a thickness of 450 Å thereon to form an electron blocking layer.

[0165] Then, the organic alloy 1 prepared in Preparation Example 3-1 and the compound GD as a dopant material were vacuum-deposited in a weight ratio of 92:8 to a thickness of 350 Å on the electron blocking layer to form a light emitting layer.

[0166] Then, the following ET-A was vacuum-deposited to a thickness of 50 Å as a hole blocking layer. Then, the following ET-B and the following Liq were thermally vacuum-deposited in a ratio of 1:1 to a thickness of 300 Å as an electron transport layer, and then Yb (ytterbium) was vacuum-deposited to a thickness of 10 Å as an electron injection layer.

[0167] Magnesium and silver were deposited in a ratio of 1:4 to a thickness of 150 Å on the electron injection layer to form a cathode, thereby completing the manufacture of an organic light emitting device.

##STR00240##

[0168] In the above-mentioned process, the vapor deposition rate of the organic material was maintained at 0.4~0.7 Å/sec, the deposition rate of magnesium and silver was maintained at 2 Å/sec, and the degree of vacuum during the deposition was maintained at 2×10^{-6} to 7.5×10^{-6} torr, thereby manufacturing an organic light emitting device.

Examples 2 to 22, and Comparative Examples 1-1 to 2-4

[0169] The organic light emitting devices of Examples 2 to 22 and Comparative Examples 1-1 to 2-4 were manufactured in the same manner as in Example 1, except that the host material was changed as shown in Table 2 below. Wherein, in Examples 19 to 22 and Comparative Examples 2-1 to 2-4, a simple mixture of two types of compounds was used as a host.

Experimental Example

[0170] The organic light emitting devices manufactured in Examples 1 to 22 and Comparative Examples 1-1 to 2-4 were heat-treated in an oven at 120° C. for 30 minutes, then taken out, and the voltage, efficiency, and lifetime (T95) were measured by applying a current, and the results are shown in Table 2 below.

[0171] At this time, the driving voltage and efficiency were measured by applying a current density of 10 mA/cm.sup.2, and T95 means the time (hr) required for the luminance to be reduced to 95% of the initial luminance at a current density of 20 mA/cm.sup.2.

TABLE-US-00002 TABLE 2 Efficiency Lifetime Voltage (V) (cd/A)@ (T95, hr) Host material @ 10 mA/cm.sup.2 10 mA/cm.sup.2 @ 20 mA/cm.sup.2 Example 1 Organic alloy 1 4.13 73.9 169 Example 2 Organic alloy 2 4.19 72.4 169 Example 3 Organic alloy 3 4.08 72.9 168 Example 4 Organic alloy 4 4.02 72.1 165 Example 5 Organic alloy 5 4.08 73.6 172 Example 6 Organic alloy 6 4.03 72.5 171 Example 7 Organic alloy 7 4.04 71.6 168 Example 8 Organic alloy 8 4.05 73.9 163 Example 9 Organic alloy 9 4.16 71.1 166 Example 10 Organic alloy 10 4.15 71.2 167 Example 11 Organic alloy 11 4.04 71.8 166 Example 12 Organic alloy 12 4.14 71.2 179 Example 13 Organic alloy 13 4.05 73.6 175 Example 14 Organic alloy 14 4.04 72.6 168 Example 15 Organic alloy 15 4.1 73.1 156 Example 16 Organic alloy 16 4.13 71.4 176 Example 17 Organic alloy 17 4.18 71.8 168 Example 18 Organic alloy 18 4.13 73.9 160 Example 19 Simple mixture of 4.13 70.8 147 Compound 1-1: Compound 2-1 in a weight ratio of 40:60 Example 20 Simple mixture of 4.19 70.7 152 Compound 1-2: Compound 2-2 in a weight ratio of 40:60 Example 21 Simple mixture of 4.05 70.5 148 Compound 1-5: Compound 2-4 in a weight ratio of 40:60 Example 22 Simple mixture of 4.16 70 149 Compound 1-6: Compound 2-5 in a weight ratio of 40:60 Comparative Organic alloy A 5.05 58.8 89 Example 1-1 Comparative Organic alloy B 4.43 60.5 89 Example 1-2 Comparative Organic alloy C 4.89 59.1 95 Example 1-3 Comparative Organic alloy D 4.21 62.7 100 Example 1-4 Comparative Compound H-1: 5.05 57.4 81 Example Compound H-2 in a 2-1 weight ratio of 40:60 Comparative Compound H-1: 4.43 58.9 84 Example Compound H-3 in a 2-2 weight ratio of 40:60 Comparative Compound 1-1: 4.89 58.8 83 Example Compound H-2 in a 2-3 weight ratio of 40:60 Comparative Compound 1-1: 4.21 61.2 91 Example Compound H-3 in a 2-4 weight ratio of 40:60

[0172] Referring to Table 2, it can be confirmed that when the compound represented by Chemical Formula 1 and the compound represented by Chemical Formula 2 are used as a host for the light emitting layer of an organic light emitting device, it has low voltage and high efficiency characteristics, and has excellent lifetime characteristics, as compared with the compounds used in Comparative Examples. In addition, it can be confirmed that when the organic alloy prepared from the compound represented by Chemical Formula 1 and the compound represented by Chemical Formula 2 is used, the efficiency and lifetime are further increased as compared with when the compounds of Chemical Formula 1 and Chemical Formula 2 are simply mixed and used.

DESCRIPTION OF REFERENCE NUMERALS

[0173] **1**: substrate [0174] **2**: anode [0175] **3**: light emitting layer [0176] **4**: cathode [0177] **5**: hole injection layer [0178] **6**: hole transport layer [0179] **7**: electron transport layer [0180] **8**: electron injection layer [0181] **9**: electron blocking layer [0182] **10**: hole blocking layer

Claims

1. An organic light emitting device comprising: an anode; a cathode; and a light emitting layer between the anode and the cathode, wherein the light emitting layer includes a compound represented by the following Chemical Formula 1 and a compound represented by the following Chemical Formula 2. ##STR00241## in Chemical Formula 1, Y is O or S, X.sub.1 to X.sub.3 are each independently CH or N, with the proviso that at least one of X.sub.1 to X.sub.3 is N, Ar.sub.1 and Ar.sub.2 are each independently a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting

of N, O and S, n is an integer of 1 to 6, R.sub.1 is each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting of N, O and S, and Ar.sub.3 is a substituent represented by the following Chemical Formula 1-1 or 1-2, ##STR00242## in Chemical Formulas 1-1 and 1-2, p is an integer of 1 to 8, q is an integer of 1 to 10, R.sub.11 and R.sub.12 are each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting of N, O and S, A is a benzene ring, and Ar.sub.12 is a substituted or unsubstituted C.sub.6-60 aryl, ##STR00243## in Chemical Formula 2, n' and m' are each independently an integer from 1 to 7, R'.sub.1 and R'.sub.2 are each independently hydrogen; deuterium; a substituted or unsubstituted C.sub.6-60 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing one or more selected from the group consisting of N, O and S, and Ar'.sub.1 and Ar'.sub.2 are each independently a substituted or unsubstituted C.sub.6-17 aryl; or a substituted or unsubstituted C.sub.2-60 heteroaryl containing any one or more heteroatoms selected from the group consisting of N, O and S, with the proviso that at least one of R'.sub.1 and R'.sub.2 is deuterium; and/or at least one of Ar'.sub.1 and Ar'.sub.2 is substituted with at least one or more deuteriums.

2. The organic light emitting device according to claim 1, wherein X.sub.1 to X.sub.3 are all N.
3. The organic light emitting device according to claim 1, wherein Ar.sub.1 and Ar.sub.2 are each independently phenyl; phenyl substituted with 5 deuteriums; biphenyl; biphenyl substituted with 5 deuteriums; carbazolyl; dibenzofuranyl; or dibenzothiophenyl.
4. The organic light emitting device according to claim 1, wherein each R.sub.1 is independently hydrogen or deuterium.
5. The organic light emitting device according to claim 1, wherein R.sub.11 is each independently hydrogen; deuterium; or phenyl unsubstituted or substituted with 5 deuteriums.
6. The organic light emitting device according to claim 1, wherein the substituent of Chemical Formula 1-2 is represented by the following Chemical Formula 1-2-a: ##STR00244## in Chemical Formula 1-2-a, q, R.sub.12, and Ar.sub.12 are the same as defined in claim 1.
7. The organic light emitting device according to claim 1, wherein R.sub.12 is each independently hydrogen or deuterium, and Ar.sub.12 is phenyl.
8. The organic light emitting device according to claim 1, wherein at least one R.sub.1 is deuterium; and/or at least one of Ar.sub.1 to Ar.sub.3 is substituted with one or more deuteriums.
9. The organic light emitting device according to claim 1, wherein Ar.sub.3 is a substituent represented by Chemical Formula 1-1, p is an integer from 4 to 8, and R.sub.11 is deuterium.
10. The organic light emitting device according to claim 1, wherein the compound represented by Chemical Formula 1 is one selected from the following: ##STR00245## ##STR00246## ##STR00247## ##STR00248## ##STR00249## ##STR00250## ##STR00251## ##STR00252## ##STR00253## ##STR00254## ##STR00255## ##STR00256## ##STR00257## ##STR00258## ##STR00259## ##STR00260## ##STR00261## ##STR00262## ##STR00263## ##STR00264## ##STR00265## ##STR00266## ##STR00267## ##STR00268## ##STR00269## ##STR00270## ##STR00271## ##STR00272## ##STR00273## ##STR00274## ##STR00275## ##STR00276## ##STR00277## ##STR00278## ##STR00279## ##STR00280## ##STR00281## ##STR00282## ##STR00283## ##STR00284## ##STR00285## ##STR00286## ##STR00287## ##STR00288## ##STR00289## ##STR00290## ##STR00291## ##STR00292## ##STR00293## ##STR00294## ##STR00295## ##STR00296## ##STR00297## ##STR00298## ##STR00299## ##STR00300## ##STR00301## ##STR00302## ##STR00303## ##STR00304## ##STR00305## ##STR00306## ##STR00307## ##STR00308## ##STR00309## ##STR00310## ##STR00311## ##STR00312## ##STR00313## ##STR00314## ##STR00315## ##STR00316## ##STR00317## ##STR00318## ##STR00319## ##STR00320## ##STR00321## ##STR00322## ##STR00323## ##STR00324## ##STR00325## ##STR00326## ##STR00327## ##STR00328## ##STR00329## ##STR00330##

##STR00331## ##STR00332## ##STR00333## ##STR00334## ##STR00335## ##STR00336##
##STR00337## ##STR00338## ##STR00339## ##STR00340## ##STR00341## ##STR00342##
##STR00343## ##STR00344## ##STR00345## ##STR00346## ##STR00347## ##STR00348##
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##STR00397## ##STR00398## ##STR00399## ##STR00400##

11. The organic light emitting device according to claim 1, wherein R'.sub.1 and R'.sub.2 are each independently hydrogen; deuterium; or phenyl unsubstituted or substituted with 1 to 5 deuteriums.

12. The organic light emitting device according to claim 1, wherein n' and m' are each independently an integer of 6 to 7, and both R'.sub.1 and R'.sub.2 are deuterium.

13. The organic light emitting device according to claim 1, wherein Ar'.sub.1 and Ar'.sub.2 are each independently phenyl unsubstituted or substituted with 1 to 5 deuteriums; biphenyl unsubstituted or substituted with 1 to 9 deuteriums; naphthyl unsubstituted or substituted with 1 to 7 deuteriums; dimethylfluorenyl unsubstituted or substituted with 1 to 13 deuteriums; dibenzofuranyl unsubstituted or substituted with 1 to 7 deuterium; or dibenzothiophenyl unsubstituted or substituted with 1 to 7 deuteriums.

14. The organic light emitting device according to claim 1, wherein the compound represented by Chemical Formula 2 is one selected from the following: ##STR00401## ##STR00402##
##STR00403## ##STR00404## ##STR00405## ##STR00406## ##STR00407## ##STR00408##
##STR00409## ##STR00410## wherein methyl groups included in each of the above listed compounds including the same are each independently CH.sub.3, CH.sub.2D, CHD.sub.2, or CD.sub.3.

15. The organic light emitting device according to claim 1, wherein the light emitting layer comprises an organic alloy of the compound represented by Chemical Formula 1 and the compound represented by Chemical Formula 2.
